You are an automated program repair tool.

Code Introduction (1)
Here is the complete test code:
{Test Code}
Please identify which test cases are used to test boundary conditions and group the test cases accordingly.

(a) The prompt for Ciallo to talk to the LLMs for the first time

You are an automated program repair tool.

Code Introduction (1)
Here is the complete test code:
{Test Code}
Error Trace (2)
Please identify which test cases are used to test boundary conditions and group the test cases accordingly.

Response of LLMs

Now you understand the scenarios these test cases are targeting.
Bug Details (3)

Now you understand the root cause of the bug. If you were to fix this buggy code now, from which directions might you approach the repair? Please list all possible directions in numbered order.
Here is an example output format. Please respond in the same format (each direction starts with a number and is entirely enclosed in **):
An example of output format...

(b) The prompt for Ciallo to talk to the LLMs for the second time

You are an automated program repair tool.

Code Introduction (1)
Here is the complete test code:
{Test Code}
Please identify which test cases are used to test boundary conditions and group the test cases accordingly.

Response of LLMs

Now you understand the scenarios these test cases are targeting.
Bug Details (3)

Please follow the format requirements above and perform repairs according to the direction: {Direction}. Please be careful not to introduce new variables or functions.
An example of output format...

(c) The prompt for Ciallo to talk to the LLMs for the third time

You are an automated program repair tool.

Code Introduction (1)
Bug Details (3)
Action Commands (4)

Patches from the previous iteration

The fixed version is still not correct. ***Feedback (5)***
Global Information (6)
Action Commands (4)

(d) The prompt for Ciallo to talk to the LLMs during patch generation iterations

You are an automated program repair tool.

Code Introduction (1)
Bug Details (3)
Solution Message (7)

Action Commands (8)

(e) The prompt for Ciallo to talk to the LLMs during patch multiplication iterations

(1)
*SL:*The following code contains a buggy line that has been removed.{Code Line}
*SH:*The following code contains a buggy hunk that has been removed.{Code Hunk}
*SF:*The following code contains a bug.{Code Function}
*BF:*The following code contains a bug.{Code File}

(2)
*SF/BF:*Here is the corresponding code in the call stack of the error message:{Context Code}
//NOTE: This part of the prompt is only added if there are lines of source code in the function call stack, and the content of the Context Code is a total of five lines of source code before and after the indicated behavior center.

(3)
*SL:*This was the original buggy line which was removed by the infill location:{Code Line}
*SH:*This was the original buggy hunk which was removed by the infill location:{Code Hunk}
The code fails on this test:{Test Name}
on this test line:{Test Line}
with the following test error:{Test Error Message}

(4)
*SL:*Please provide the correct line at the infill location.
*SH:*Please provide the correct hunk at the infill location.
*SF:*Please provide the correct function.
*BF:*Please provide the correct program.

(5)
When the error result is same as the original: It still does not fix the original test failure.
*When the error result is different from the original:*Code has the following {Error Type}: {Reason}

(6)
*SL/SH/SF:*Here is the global variables and methods, you can use them to help you to repair the buggy code:{Global Information}
//NOTE: This part of the prompt is only added if the error type is "cannot find symbol"

(7)
*SL:*It can be fixed by these possible line:
*SH:*It can be fixed by the following hunk:
*SF:*It can be fixed by the following function:
*BF:*It can be fixed by the following program:

(8)
*SL:*Please generate an alternative fix line.
*SH:*Please generate an alternative fix hunk.
*SF:*Please generate an alternative fix function.
*BF:*Please generate an alternative fix program.

(f) Details of prompts